

In this section we consider series that include powers of a variable x , for example:

$$\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + \dots$$

where $c_0, c_1, \dots$ are constants.

Partial sum S_n of a power series is a polynomial of degree n

Example: Use long division to find a series expression for $f(x) = \frac{1}{1-x}$.

$$f(x) = \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

↳ this is a function that is $\Rightarrow$ to a power series

↳ Geometric

Series w/ $r=x$, $a=1$. Converges for $|x| < 1$

Not true for every value of x . Only true

So long as the series converges

So long as the series converges

Important Question: For what values of x does $\sum_{n=0}^{\infty} c_n x^n$ converge?

• If $x=0$, $\sum_{n=0}^{\infty} c_n x^n = c_0$ $c_0 + \cancel{c_1 x} + \cancel{c_2 x^2} + \dots$

• If $x \neq 0$, apply ratio test only applicable test for power series

$$\text{Let } a_n = c_n x^n \text{ (so } \sum_{n=0}^{\infty} c_n x^n = \sum_{n=0}^{\infty} a_n) \rightarrow \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{c_{n+1} x^{n+1}}{c_n x^n} \right| = |x| \lim_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right| = L|x|$$

If $L = \infty$, then the only term that converges is c_0 .

If $L|x| < 1$, power series converges

If $L|x| > 1$, power series diverges

If $L|x| = 1$, inconclusive

$$\text{Let } L = \lim_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right|$$

$$\sum_{n=0}^{\infty} c_n x^n \begin{cases} \text{Converges for } |x| < \frac{1}{L} \\ \text{diverges for } |x| > \frac{1}{L} \end{cases}$$

$$L = \lim_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right|$$

plug in $x = \pm \frac{1}{L}$ and test series

$\frac{1}{L}$ is called the **radius of convergence**. When $|x| = \frac{1}{L}$, test endpoints manually

Ratio test used to get ROC

Example: Find the radius and interval of convergence for: $\sum_{n=1}^{\infty} \frac{x^n}{n 3^n}$

$$C_n = \frac{1}{n 3^n}$$

$$L = \lim_{n \rightarrow \infty} \left| \frac{C_{n+1}}{C_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{1}{(n+1) 3^{n+1}} \cdot \frac{n 3^n}{1} \right| = \frac{1}{3} \lim_{n \rightarrow \infty} \left| \frac{n}{n+1} \right| = \frac{1}{3}$$

$$L = \frac{1}{3} \rightarrow \text{radius is } 3$$



Now,

- If $x=3$, $\sum_{n=1}^{\infty} \frac{3^n}{n 3^n} = \sum_{n=1}^{\infty} \frac{1}{n}$ *diverges* $\rightarrow$ *Excluded*
harmonic series

- If $x=-3$, $\sum_{n=1}^{\infty} \frac{(-3)^n}{n 3^n} = \sum_{n=1}^{\infty} (-1)^n \frac{1}{n}$ *Converges* $\rightarrow$ *Included*
Alternating harmonic series

So $\sum_{n=1}^{\infty} \frac{x^n}{n 3^n}$ converges for $-3 \leq x < 3$
 or
 $[-3, 3)$

Example: Find the radius and interval of convergence for:

starting index
does not matter

$$\sum_{n=0}^{\infty} \frac{2^n x^n}{n!}$$

$$C_n = \frac{2^n}{n!} \rightarrow C_{n+2} = \frac{2^{n+2}}{(n+2)n!}$$

$$L = \lim_{n \rightarrow \infty} \left| \frac{2^{n+2}}{(n+2)n!} \cdot \frac{n!}{2^n} \right| = \lim_{n \rightarrow \infty} \left| \frac{2}{n+2} \right| = 0$$

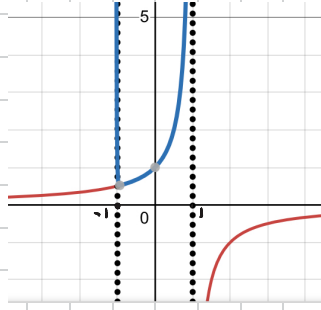
If $L = 0$, radius of convergence = ∞
 $\hookrightarrow$ Converges regardless of value of x

$L|x| = 0|x| = 0$ regardless
 of value of
 x , and $0 < L$
 so it always
 is < 1 regard-
 less of value
 of x

So $\sum_{n=0}^{\infty} \frac{2^n x^n}{n!}$ converges for all x or $(-\infty, \infty)$

You don't
 need to test
 endpoints here

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad \text{So long as } |x| < 1$$



Within this radius,

$$\sum_{n=0}^{100} x^n \text{ behaves like}$$

$$\frac{1}{1-x} \text{ within } \pm 1$$

$\sum_{n=0}^{\infty} c_n x^n$ is a power series centered at $x=0$

More generally,

$\sum_{n=0}^{\infty} c_n (x-a)^n$ is a power series but centered at $x=a$

Power series in $(x - a)$:

$$\sum_{n=0}^{\infty} c_n(x-a)^n$$

works
regardless
of where
series
is centered

where $c_0, c_1, \dots$ and a are constants.

To find the radius and interval of convergence:

- Again use ratio test to find: $\lim_{n \rightarrow \infty} \left| \frac{c_{n+1}(x-a)^{n+1}}{c_n(x-a)^n} \right| = L |x-a|$
- Converges if $L|x-a| < 1$
Diverges if $L|x-a| > 1$
Inconclusive if $L|x-a| = 1$

Example: Find the radius and interval of convergence for:

$$\sum_{n=1}^{\infty} (-1)^n \frac{(x-2)^n}{4^n \sqrt{n}}$$

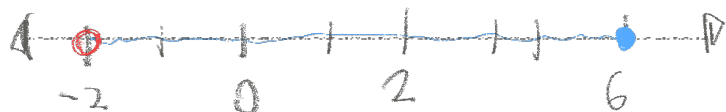
$$C_n = \frac{(-2)^n}{4^n \sqrt{n}}$$

Everything that doesn't depend on x

$$\sum_{n=1}^{\infty} C_n (x-2)^n \quad \text{power series centered at } x=2$$

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} \left| \frac{C_{n+1}}{C_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{\cancel{(-2)^{n+1}} \cdot (-1)}{\cancel{4^{n+1}} \sqrt{n+1}} \cdot \frac{\cancel{4^n} \sqrt{n}}{\cancel{(-2)^n}} \right| = \lim_{n \rightarrow \infty} \left| \frac{(-2) \sqrt{n}}{4 \sqrt{n+1}} \right| \\ &= \lim_{n \rightarrow \infty} \frac{\sqrt{n}}{4 \sqrt{n+1}} \\ &= \frac{1}{4} \end{aligned}$$

So radius of convergence is 4, $\rightarrow (4|x-2| < 4)$



End points are -2 and 6. Now test them

• If $x = -2$, $\sum_{n=1}^{\infty} (-1)^n \frac{(-4)^n}{4^n \sqrt{n}} = \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} \rightarrow$ Diverges as p -series ($p = \frac{1}{2} < 1$)

Why does it cancel like this

• If $x = 6$, $\sum_{n=1}^{\infty} (-1)^n \frac{4^n}{4^n \sqrt{n}} = \sum_{n=1}^{\infty} (-1)^n \frac{1}{\sqrt{n}} \rightarrow$ Converges by AST

Converges for $-2 < x \leq 6$